

SKILLS AND COMPETENCIES

LLMs, DL, & ML: PyTorch, Huggingface, Transformers, Accelerate, AWS SageMaker, fastai, Ax, Optuna, scikit-learn, pandas, numpy, spaCy.

ML Engineering: Experimentation, Fine-tuning, Evaluation pipelines, Predictive modeling, Scalable deployment.

MLOps, DevOps, CI/CD: MLFlow, Neptune, LightningAI, AWS, GCP, CI/CD pipelines, Docker, Kubernetes, Prometheus, Grafana.

Full-Stack Engineering: FastAPI, OpenAPI, REST APIs, Microservices for inference, Streamlit, D3.js, Test Driven Development (TDD), Front-end frameworks, Back-end frameworks.

Collaboration & Leadership: Mentorship, Code reviews, Agile/Scrum, Research-to-production translation, Strong verbal and written communication skills, Strong work ethic, Self-starter, Problem solver.

EXPERIENCE

R&D Scientist II - Quantum NLP (QNLP)

Quantinuum Ltd.

Oct 2023 — Present

Oxford, United Kingdom

- **LLM Fine-tuning for Structured Reasoning Tasks**

- Designed and deployed LLM fine-tuning pipelines (SFT+GRPO), utilizing PEFT and multi-GPU training with unsloth and LightningAI to train Qwen2 models for the task of creating and manipulating string diagrams that required formally verifiable steps or Chain-of-Thought (CoT) reasoning.
- Worked with senior researchers to translate their domain knowledge in string diagrams and category theory into synthetic data generation scripts, serialization/deserialization functions, and instruction prompts for creating the training and evaluation datasets.
- Designed custom evaluation metrics to measure the performance of the trained models.

- **Novel BERT model for End-to-End Generation of String Diagrams from Sentences**

- Designed, implemented, and trained a custom BERT-based model for converting sentences directly to string diagrams which simplified QNLP training pipelines , utilizing LoRA fine-tuning, automated hyperparameter tuning using Ax, and experiment and artifact tracking using both Neptune and MLFlow for more efficiency and reproducibility.
- Created a compact tree-based representation of string diagrams to create the training and evaluation datasets and an appropriate data augmentation strategy for the task.
- Worked closely with interns, allowing them to contribute to the evaluation methods, while maintaining good codebase quality for ease of reproducibility, maintainability, and extensibility.

- **Translation of Theoretical Research to Practical Implementations**

- Collaborated with senior researchers on an improved implementation of the DisCoCirc paper  which describes a compositional, non-LLM-based model for how texts derive meaning from their constituent sentences, within a tight timeline of 2 months.
- Created benchmarking datasets to compare the performance of the new implementation versus the original implementation  and wrote reusable, configurable scripts to check the coverage of the new implementation on the dataset which simplified the testing pipeline and increased the confidence with the new pipeline.
- Implemented visualization functions to allow inspection of the output circuit diagrams.
- Presented work to other colleagues not necessarily familiar with QNLP.

- **lambeq Open-Source Library Research and Development**

- Implemented new features and bug fixes for better usability when training QNLP models.
- Streamlined APIs, removed bottlenecks, improved documentation, and created example notebooks to onboard new users.
- Mentored interns and led regular code and PR reviews with the team to enforce best practices.
- Replicated and built upon the results of past research papers for miscellaneous areas of interest.

Data Scientist

AI Cross Inc.

Nov 2021 — Sep 2023

Tokyo, Japan

- **Machine Learning Platform-as-a-Service** 🔗

- Designed and implemented the core ML module of the platform that allowed users to train models and retrieve inference results via APIs.

- **Business Marketing Platform-as-a-Service** 🔗

- Developed prototype for generating and delivering AI-based recommendations in 1 month, which enabled the team to pitch and to present to potential clients, and to validate the value provided by the service.
- Created the system architecture for the product based on the initial prototype.
- Implemented the data ingestion, model training, and recommender components of the system, working across the entire stack (frontend, backend, ML module).
- Constantly communicated with the sales team to understand client needs and re-prioritized the development milestones accordingly.
- Implemented A/B testing feature that allowed clients to experiment with their strategies and choose the best option within statistical bounds.
- Self-learned about UI/UX and implemented changes to the design of the platform.
- Operationalized the system, closely collaborating with our CTO in the process.

Machine Learning Engineer

LINE Fukuoka Corporation

Sep 2019 — Nov 2021

Fukuoka, Japan

- **Detection of High-Profile Individuals from Caricatures**

- Built a high-throughput (1M+/sec) image recognition pipeline based on the WarpGAN paper using Spark and Airflow.
- Developed and deployed APIs for real-time inference and visualization of batch detection results with service monitoring via Prometheus and Grafana to tighten the feedback loop in detecting target events and implemented CI/CD pipelines for reliable deployment.

- **Content Suggestion via Image-Based Emotion Recognition**

- Designed the model and system architecture for a new emotion recognition-based recommendation system feature to the LINE messaging app, collaborating closely with the product team.
- Developed and deployed APIs for real-time inference complete with service monitoring via Prometheus and Grafana.

- **Machine Learning Tooling and Knowledge-Sharing Tasks**

- Improved existing and created new ML/MLOps tools with the team.
- Gave presentations on the following areas of interest: meta-learning, model quantization 🔗, auto-ML 🔗, causal inference 🔗, secure AI 🔗, and explainable AI 🔗.

Data Scientist

Teradata Corporation

Nov 2016 — Sep 2019

Manila, Philippines

- **Survey Response Classification**

- Developed NLP models (LSTM + embedding) using Keras with Tensorflow and scikit-learn for the classification of survey responses, improving automation.
- Designed end-to-end (E2E) pipelines with SQL ingestion, Dockerized deployment, and prediction visualization that reduced manual labor and errors made by client.
- Led knowledge-sharing sessions for new technologies for the rest of the team.

Senior Full-Stack Engineer

icannhas Inc.

Sep 2011 — Nov 2016

Manila, Philippines

EDUCATION

Master of Science in Physics

University of the Philippines Diliman, Quezon City (On leave)

2021 — Present

Bachelor of Science in Civil Engineering, magna cum laude

University of the Philippines Diliman, Quezon City

2006 — 2011

Awardee, Engineering Top 100 (out of 5000 students) in Academics, 2011

10th Place, ACM International Collegiate Programming Contest Asia Manila Regional Contest, 2009

Honorable Mention, 37th International Physics Olympiad, 2006